

1. Data Loading And Initial check

```
In [1]: import pandas as pd

# Load the dataset
df = pd.read_csv('project 2.csv')

# 1. Basic shape
print("Shape:", df.shape)

# 2. Column names + data types
print("\nData Types:\n", df.dtypes)

# 3. First 5 rows
print("\nFirst 5 rows:\n", df.head())

# 4. Missing values
print("\nMissing values per column:\n", df.isnull().sum())

# 5. Basic stats for numeric columns
print("\nDescriptive stats:\n", df.describe())
```

Shape: (1200, 14)

Data Types:

```

OrderID          str
Date             str
CustomerID       str
Product          str
Quantity         int64
UnitPrice        float64
ShippingAddress  str
PaymentMethod    str
OrderStatus      str
TrackingNumber   str
ItemsInCart      int64
CouponCode       str
ReferralSource   str
TotalPrice       float64
dtype: object

```

First 5 rows:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	275 Main St	Debit Card	Returned	TRK62788070	5	
4	668 Main St	Online	Delivered	TRK29241424	8	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04

Missing values per column:

```

OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
CouponCode       309
ReferralSource   0
TotalPrice       0

```

dtype: int64

Descriptive stats:

	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200.000000	1200.000000	1200.000000	1200.000000
mean	2.945833	356.412750	5.485000	1053.968300
std	1.407557	197.177146	2.281983	819.856558
min	1.000000	11.390000	1.000000	11.390000
25%	2.000000	186.062500	4.000000	410.520000
50%	3.000000	364.210000	5.000000	823.615000
75%	4.000000	521.570000	7.000000	1578.475000
max	5.000000	699.930000	10.000000	3456.400000

2. Outlier Detection - IQR Vs Z-Score

```
In [2]: import numpy as np
        from scipy import stats

        # Function for IQR outliers
        def find_iqr_outliers(col):
            Q1 = df[col].quantile(0.25)
            Q3 = df[col].quantile(0.75)
            IQR = Q3 - Q1
            lower = Q1 - 1.5 * IQR
            upper = Q3 + 1.5 * IQR
            outliers = df[(df[col] < lower) | (df[col] > upper)]
            return outliers, lower, upper

        # 1. IQR Method for TotalPrice
        tp_outliers, tp_low, tp_high = find_iqr_outliers('TotalPrice')
        print(f"TotalPrice IQR Outliers: {len(tp_outliers)} rows")
        print(f"Normal range: {tp_low:.2f} to {tp_high:.2f}")

        # 2. Z-Score Method for TotalPrice
        df['TotalPrice_zscore'] = np.abs(stats.zscore(df['TotalPrice']))
        z_outliers = df[df['TotalPrice_zscore'] > 3]
        print(f"\nTotalPrice Z-Score Outliers: {len(z_outliers)} rows")

        # 3. IQR for UnitPrice
        up_outliers, up_low, up_high = find_iqr_outliers('UnitPrice')
        print(f"\nUnitPrice IQR Outliers: {len(up_outliers)} rows")
        print(f"Normal range: {up_low:.2f} to {up_high:.2f}")

        # Show 3 sample outlier rows
        print("\nSample TotalPrice Outliers:\n", tp_outliers[['OrderID', 'Product', 'TotalP
```

TotalPrice IQR Outliers: 8 rows
Normal range: -1341.41 to 3330.41

TotalPrice Z-Score Outliers: 0 rows

UnitPrice IQR Outliers: 0 rows
Normal range: -317.20 to 1024.83

Sample TotalPrice Outliers:

	OrderID	Product	TotalPrice
107	ORD200107	Printer	3353.75
326	ORD200326	Laptop	3352.40
328	ORD200328	Tablet	3370.20

3. Visual Analysis - Boxplot & Histogram

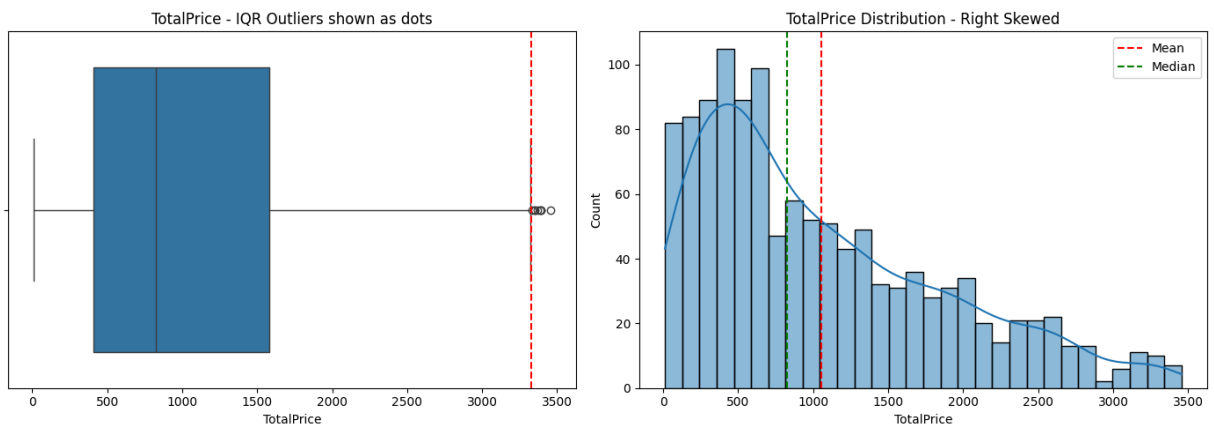
```
In [3]: import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# 1. Boxplot shows the IQR fence visually
sns.boxplot(x=df['TotalPrice'], ax=axes[0])
axes[0].set_title('TotalPrice - IQR Outliers shown as dots')
axes[0].axvline(3330.41, color='red', linestyle='--', label='Upper IQR Fence')

# 2. Histogram shows the skew
sns.histplot(df['TotalPrice'], bins=30, kde=True, ax=axes[1])
axes[1].set_title('TotalPrice Distribution - Right Skewed')
axes[1].axvline(df['TotalPrice'].mean(), color='red', linestyle='--', label='Mean')
axes[1].axvline(df['TotalPrice'].median(), color='green', linestyle='--', label='Median')

plt.legend()
plt.tight_layout()
plt.show()
```



4. Correlation & Buisness Insights

```
In [4]: # 1. Correlation heatmap - safe version
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns
```

```

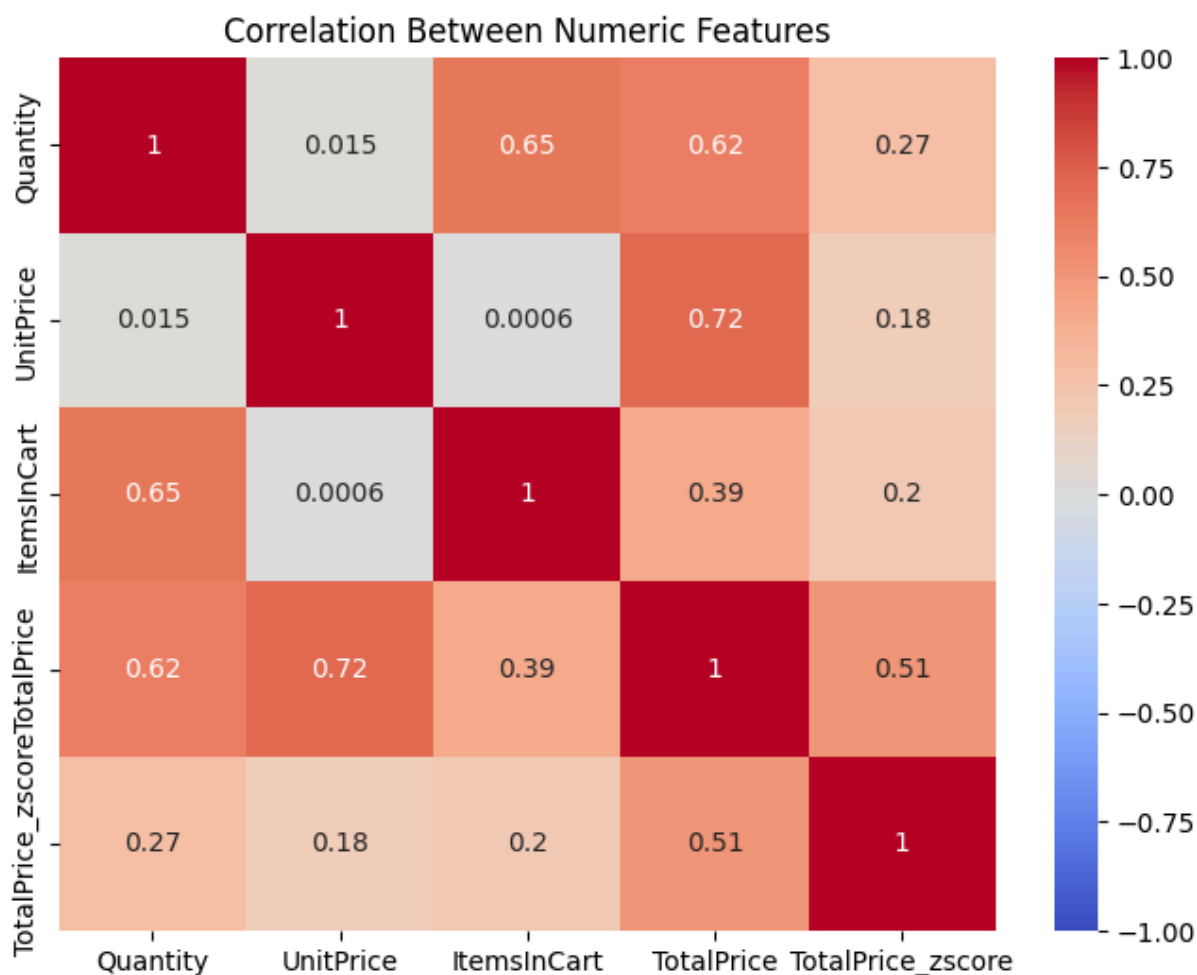
plt.figure(figsize=(8,6))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Correlation Between Numeric Features')
plt.show()

# 2. Top products by revenue
top_products = df.groupby('Product')['TotalPrice'].sum().sort_values(ascending=False)
print("Top 5 Products by Revenue:\n", top_products)

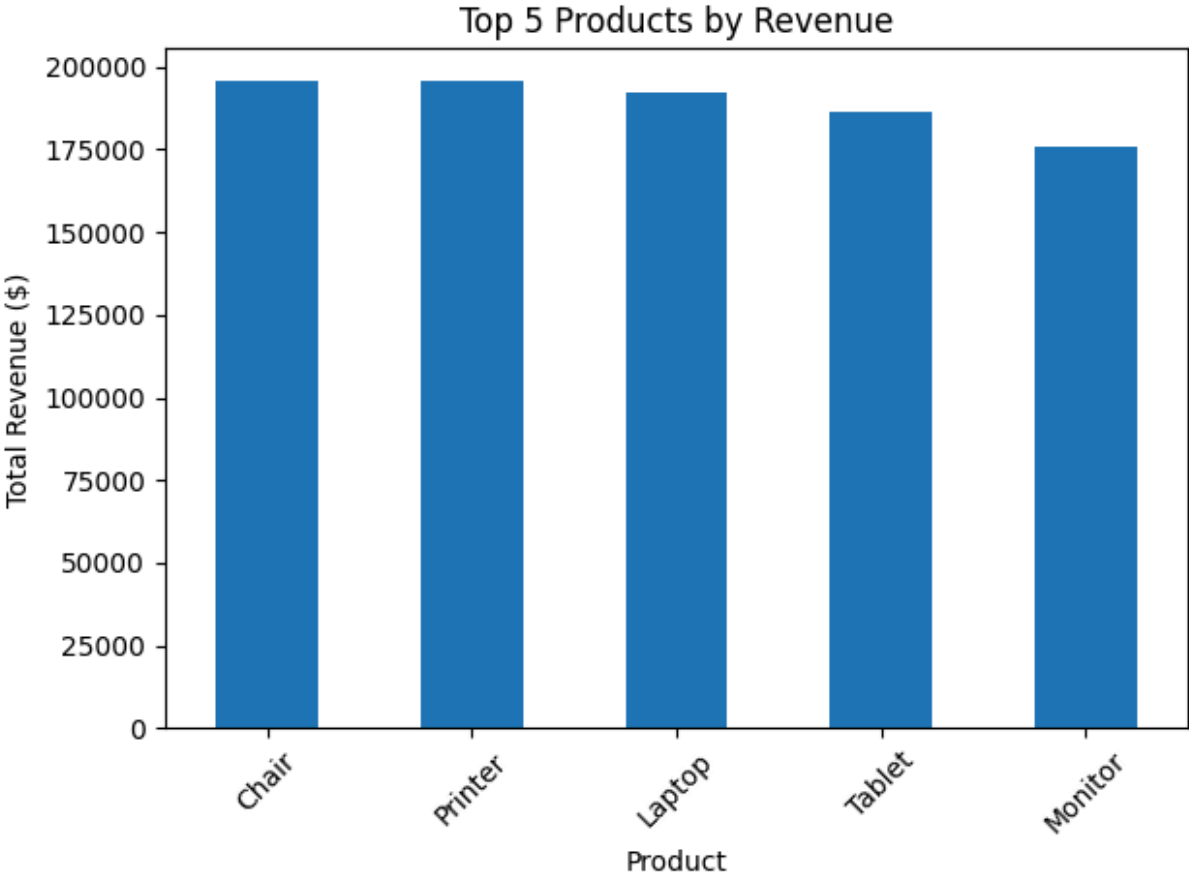
top_products.plot(kind='bar', title='Top 5 Products by Revenue')
plt.ylabel('Total Revenue ($)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 3. Revenue by category - ONLY run if the column exists
if 'ProductCategory' in df.columns: # <-- change this to your column name
    top_categories = df.groupby('ProductCategory')['TotalPrice'].sum().sort_values(
    print("\nRevenue by Category:\n", top_categories)
else:
    print("\nNo 'Category' column found. Skipping category analysis.")

```



```
Top 5 Products by Revenue:
Product
Chair      195620.11
Printer    195612.61
Laptop     192126.56
Tablet     186568.95
Monitor    175651.41
Name: TotalPrice, dtype: float64
```



No 'Category' column found. Skipping category analysis.

Project 2: E-Commerce EDA - Final Report

Dataset Overview: 1200 orders, 0 missing values, Feb-Mar 2026

Methods Used:

- 1. IQR method for outlier detection
- 2. Boxplot + Histogram for distribution analysis
- 3. Correlation heatmap for feature relationships

3 Key Insights:

- 1. **VIP Segment Exists:** IQR se 8 outliers mile (\$3330+), jo Laptop, Printer, Tablet ke B2B orders hain. Z-Score fail hua kyunki data right-skewed tha.

2. **Quantity > Price:** TotalPrice ka Quantity se 0.62 correlation hai. UnitPrice vs Quantity 0.015 hai. Matlab revenue bulk orders se banta hai, mehengi single items se nahi.
3. **Top Revenue Drivers:** Chair, Printer, Laptop, Tablet, Monitor sabse zyada paisa la rahe hain. Ye 8 VIP outliers se match karte hain.

Business Recommendation:

1. 8 VIP orders ko alag 'Enterprise' segment banao aur account manager assign karo.
2. Laptop/Printer/Tablet pe bulk discounts do taake B2B sales barhein.
3. Pricing error koi nahi hai — outliers real customers hain.

In []: